

# Emergence of Macroscopic Coherence based on Extended Integrated Field Equation and Predictive Error Minimization in Complex Vector Fields

Satoshi Takemoto\*

*Independent Researcher, Tokyo, Japan*

(Dated: August 28, 2026)

We extend the framework of the Nonequilibrium Integrated Phase (NEIP) from a scalar field to a complex vector field to elucidate the physical mechanisms underlying active adaptation and macroscopic coherence in complex driven systems. By incorporating inertial dynamics and variational free energy minimization into the Integrated Field Equation (IFE), we formulate a predictive error minimization dynamics governed strictly by physical principles. Furthermore, we demonstrate that the hierarchical information flow in this system can be mapped onto a renormalization group (RG) process, where external fields drive the splitting of attractors within a replica symmetry breaking (RSB)-induced multi-valley free energy landscape. This structural bifurcation provides a purely physical and geometric foundation for the compression of degrees of freedom and macroscopic synchronization, bridging the gap between statistical field theory and active information processing.

## I. INTRODUCTION

Complex nonequilibrium systems often exhibit highly coordinated behaviors, characterized by multiscale structures, history dependence, and macroscopic order. Traditional non-equilibrium statistical mechanics has primarily focused on the passive relaxation of these systems towards steady states or the characterization of phase transitions driven by thermal fluctuations and localized interactions. However, a wide class of driven macroscopic systems exhibits a more profound capacity: the ability to actively adapt to external fields and dynamically compress internal degrees of freedom. To rigorously describe such phenomena, it is imperative to bridge the gap between statistical field theory and active information processing, formulating a framework where adaptation is treated as a fundamental physical driving force.

In our previous work [1], we identified the Nonequilibrium Integrated Phase (NEIP) as a distinct regime in nonequilibrium phase space, characterized by the simultaneous emergence of integratedness, multiscaleness, temporal recurrence, and sustained dissipation. This phase was formalized through the scalar Integrated Field Equation (IFE), a minimal stochastic model incorporating gradient flow, nonlocal coupling, and non-Markovian memory [1, 2]. While the scalar IFE successfully captured the passive integration of long-range correlations and history dependence, it fundamentally lacked the phase degrees of freedom required to describe large-scale synchronization and the active minimization of external perturbations.

To overcome this limitation and capture the emergence of macroscopic coherence, this paper extends the NEIP framework from a scalar field to a complex vector field. We introduce an extended Integrated Field Equation (Vector-IFE) that incorporates inertial dynamics and a variational free energy minimization term. In

this framework, the system actively drives its phase space trajectory to minimize the generalized discrepancy—or predictive error—between its internal state and external hierarchical fields.

Furthermore, we demonstrate that the hierarchical information flow in this system can be mathematically mapped onto a renormalization group (RG) process. Under this RG flow, external fields drive the splitting of attractors within a replica symmetry breaking (RSB)-induced multi-valley free energy landscape.

By incorporating inertial dynamics and variational free energy minimization [3] into the Integrated Field Equation (IFE), we formulate a predictive error minimization dynamics [4] governed strictly by physical principles. Furthermore, we demonstrate that the hierarchical information flow in this system can be mapped onto a renormalization group (RG) process [5], where external fields drive the splitting of attractors within a replica symmetry breaking (RSB)[6]-induced multi-valley free energy landscape.

This structural bifurcation provides a purely physical and geometric foundation for the compression of degrees of freedom and macroscopic synchronization. By doing so, we establish a novel physical mechanics of active adaptation, laying the groundwork for a geometric understanding of macroscopic coherence in complex driven systems.

## II. RECONSIDERATION OF NEIP AND THE SCALAR IFE

The Nonequilibrium Integrated Phase (NEIP) was originally conceptualized to describe dynamic regimes where long-range spatial correlations, temporal memory, and sustained energy dissipation coexist in a coordinated manner. In our preceding work, we mathematically captured this phase using the scalar Integrated Field Equation (IFE). The scalar IFE combined a gradient flow in a multi-stable potential with nonlocal spatial coupling and a non-Markovian memory kernel. This formulation suc-

---

\* [s.takemoto.1944@gmail.com](mailto:s.takemoto.1944@gmail.com)

cessfully demonstrated how a physical system can exhibit a scale-dependent spectral dimension and history dependence, preventing the relaxation to a trivial equilibrium.

However, the scalar representation imposes fundamental restrictions on the macroscopic behavior of the system. A real scalar field  $\phi(x, t)$  can only describe variations in local amplitude. It lacks the continuous internal degrees of freedom necessary to represent phase dynamics. Consequently, the scalar IFE cannot capture wave propagation, interference, or the emergence of global synchronization—phenomena that are essential for large-scale coherence in highly coordinated complex systems.

Furthermore, the information integration within the scalar IFE is inherently passive. While the nonlocal kernel and memory term allow the system to retain historical states and couple distant regions, the system's trajectory is dictated solely by the pre-defined free energy landscape and stochastic forcing. It does not actively adjust its internal state to minimize generalized discrepancies against fluctuating external environments.

To describe a macroscopic field that continuously adapts its internal structure to external hierarchical fields, it is a physical necessity to introduce both phase degrees of freedom and an active driving mechanism. Table I summarizes the conceptual and mathematical transition from the passive scalar IFE to the active Vector-IFE. This geometric and dynamical expansion rigorously necessitates the transition to a complex vector field, forming the basis for the extended framework presented in this paper.

### III. FORMULATION OF THE EXTENDED VECTOR-IFE

To capture the emergence of macroscopic coherence and active adaptation, we extend the scalar field representation to a complex vector field  $\Psi(x, t)$ . In this formulation, the field can be expressed as  $\Psi(x, t) = |\Psi(x, t)|e^{i\theta(x, t)}$ , where the amplitude  $|\Psi(x, t)|$  represents the intensity of local activity, and the phase  $\theta(x, t)$  encodes the synchronization state across the spatial domain.

#### A. Inertial Dynamics and Symmetry Breaking

Highly coordinated complex systems often exhibit fast oscillatory coherence and wave-like propagation that cannot be captured by the overdamped Langevin dynamics of the previous scalar IFE. Therefore, we introduce an inertial term  $m\partial_t^2\Psi$ , where  $m$  acts as an effective mass representing inductance or transmission delays in the system, and  $\Gamma$  is the dissipation coefficient.

The baseline critical dynamics of the field is governed by the time-dependent Ginzburg-Landau potential. Incorporating these elements, the extended Inte-

grated Field Equation (Vector-IFE) is formulated as:

$$m\partial_t^2\Psi + \Gamma\partial_t\Psi = -r\Psi + u|\Psi|^2\Psi - K\nabla^2\Psi - \lambda\frac{\delta\mathcal{F}}{\delta\Psi} + \xi(x, t) \quad (1)$$

where  $\xi(x, t)$  represents Gaussian white noise. The linear coefficient  $r$  controls the spontaneous symmetry breaking of the critical field: a sign change in  $r$  drives the system from a disordered state into an ordered state with a finite amplitude, providing the physical substrate for coherence.

#### B. Non-local Integration via Small-World Topology

In addition to the local diffusive coupling represented by the Laplacian term  $-K\nabla^2\Psi$ , macroscopic integration in complex networks requires non-local interactions. In our framework, this is achieved by embedding a small-world network topology into the interaction spatial kernel.

Unlike strictly nearest-neighbor continuous limits, the small-world topology introduces long-range shortcuts that drastically reduce the characteristic path length of the system. In the context of the complex vector field  $\Psi$ , these non-local connections act to globally couple the phase  $\theta(x, t)$  across distant regions. This topological property allows the system to overcome local pinning forces and spontaneously align phases, driving the emergence of global phase synchronization (macroscopic integration) across the entire field.

#### C. Predictive Driving via Variational Free Energy

The most critical departure from standard physical field models is the introduction of the predictive driving force,  $-\lambda\frac{\delta\mathcal{F}}{\delta\Psi}$ . In purely thermodynamic systems, the free energy relies solely on the internal structural potential (e.g., the  $-r\Psi + u|\Psi|^2\Psi$  terms). However, in actively adapting systems, the field  $\Psi(x, t)$  must continuously update its state to reflect and predict fluctuating external environments.

Here,  $\mathcal{F}$  represents the variational free energy, which mathematically quantifies the generalized discrepancy (or predictive error) between the system's internal state (prior expectations) and the external fields (sensory inputs). The functional derivative  $-\frac{\delta\mathcal{F}}{\delta\Psi}$  ensures that the temporal evolution of the phase space trajectory is strictly directed toward minimizing this error. Consequently, the field  $\Psi(x, t)$  functions not merely as a passive physical substrate, but as a dynamic predictive machine that actively suppresses external surprises.

Here,  $\mathcal{F}$  represents the variational free energy [3], which mathematically quantifies the generalized discrepancy (or predictive error) between the system's internal state (prior expectations) and the external fields (sensory inputs).

| Feature / Property     | Scalar IFE (Previous)     | Vector-IFE (Current)          |
|------------------------|---------------------------|-------------------------------|
| Field Variable         | Real scalar $\phi(x, t)$  | Complex vector $\Psi(x, t)$   |
| Degrees of Freedom     | Amplitude                 | Amplitude & Phase             |
| Dynamic Equation       | Overdamped flow           | Inertial dynamics             |
| Driving Force          | Pre-defined potential     | Variational Free Energy       |
| Information Processing | Passive history retention | Predictive error minimization |
| Macroscopic Coherence  | Spatial clustering        | Phase synchronization         |
| Hierarchical Structure | Single-layer multiscaling | Multi-layer RG flow           |

TABLE I. Comparison of the physical and informational properties between the previous scalar IFE and the extended Vector-IFE frameworks.

#### IV. HIERARCHICAL FIELDS AND RENORMALIZATION GROUP CONSTRUCTION OF VARIATIONAL FREE ENERGY

To address the emergence of abstraction and multi-scale representation, we construct the variational free energy  $\mathcal{F}$  within a hierarchical field architecture. In complex driven systems, information processing does not occur at a single characteristic scale; rather, it spans a continuous hierarchy of spatial and temporal resolutions.

##### A. Hierarchical Layer Structure and Inter-Layer Coupling

We define a multi-layer field configuration  $\{\Psi^{(l)}(x, t)\}$ , where  $l = 1, 2, \dots, L$  denotes the hierarchical index. The lowest layer  $\Psi^{(1)}(x, t)$  directly interacts with fluctuating external sources, whereas higher layers  $\Psi^{(l)}$  ( $l > 1$ ) receive coarse-grained inputs from the layers immediately below.

The variational free energy  $\mathcal{F}$  governing this multi-layer structure is formulated as a sum over hierarchical discrepancies:

$$\mathcal{F} = \sum_l \int dx \left[ \frac{1}{2} \left| \Psi^{(l-1)} - \mathcal{M}^{(l)}[\Psi^{(l)}] \right|^2 + \mathcal{G}[\Psi^{(l)}] \right], \quad (2)$$

where  $\mathcal{M}^{(l)}$  is a nonlinear mapping operator representing the downward generative expectation from layer  $l$  to layer  $l - 1$ , and  $\mathcal{G}[\Psi^{(l)}]$  denotes the internal structural potential of layer  $l$ . The predictive error term  $\left| \Psi^{(l-1)} - \mathcal{M}^{(l)}[\Psi^{(l)}] \right|^2$  acts as the inter-layer coupling that drives the variational update.

The vertical integration of information across this hierarchy can be rigorously mapped onto a statistical mechanical renormalization group (RG) transformation [5]. As one ascends from layer  $l$  to layer  $l + 1$ , high-frequency spatial fluctuations and microscopic details are progressively integrated out.

##### B. Mapping to Renormalization Group (RG) Flow

The vertical integration of information across this hierarchy can be rigorously mapped onto a statistical mechanical renormalization group (RG) transformation. As one ascends from layer  $l$  to layer  $l + 1$ , high-frequency spatial fluctuations and microscopic details are progressively integrated out.

Mathematically, coarse-graining the micro-states via the hierarchical mapping corresponds to an RG blocking transformation in momentum space. The effective free energy at a macroscopic scale is obtained by scaling the coupling parameters (such as the gradient coefficient  $K$  and potential parameters  $r, u$ ). This scale transformation dictates that the compression of microscopic degrees of freedom into macroscopic variables is formally equivalent to flowing toward an infrared fixed point of the RG equations. Consequently, the “abstraction” of information—the extraction of invariant features from noisy sensory inputs—is given a precise physical foundation as a scale-invariant RG flow within the statistical manifold of the extended Vector-IFE.

## V. DYNAMICS OF MULTI-VALLEY STRUCTURE: HIGH-ORDER FORMATION AND ATTRACTOR SPLITTING

While renormalization group coarse-graining accounts for the vertical compression of microscopic degrees of freedom, the horizontal organization of internal representations requires a multi-valley free energy landscape. In complex systems, frustration and non-local interactions naturally give rise to replica symmetry breaking (RSB)-like rugged potentials characterized by a proliferation of metastable states.

### A. Spin-Glass Landscape and Multi-Valley Structure

The internal potential  $\mathcal{G}[\Psi^{(l)}]$  within each hierarchical layer exhibits a complex landscape populated by numerous local minima (valleys). Each valley corresponds to a distinct coordinated state or “prior expectation” acquired by the field. In the absence of external perturbations, the system’s phase space trajectory fluctuates among these metastable attractors due to stochastic forcing  $\xi(x, t)$ .

However, the introduction of the predictive driving force  $-\lambda \frac{\delta \mathcal{F}}{\delta \Psi}$  fundamentally alters how the system navigates this landscape. When sensory inputs or external hierarchical fields impinge upon layer  $\Psi^{(1)}(x, t)$ , the resulting predictive error tilts and deforms the local free energy landscape, forcing the phase space trajectory out of generalized traps toward states that minimize the discrepancy.

### B. Bifurcation and Attractor Splitting as Adaptive Differentiation

A profound consequence of this predictive dynamics under varying external driving is the phenomenon of *attractor splitting*. As the coupling strength or the precision of the predictive error terms increases, a single macroscopic attractor (representing an unspecialized, coarse internal state) undergoes a pitchfork-like bifurcation within the rugged landscape.

In complex systems, frustration and non-local interactions naturally give rise to replica symmetry breaking (RSB)[6]-like rugged potentials characterized by a proliferation of metastable states.

Mathematically, this corresponds to the destabilization of a symmetric saddle point in the statistical manifold, causing a primary valley to split into two or more distinct sub-valleys. In the context of active information processing, this structural bifurcation describes how a system differentiates its internal concepts or categories in response to environmental complexity. The splitting of attractors provides a rigorous physical mechanism for the emergence of high-order functional specificity—demonstrating that learning and cognitive differ-

entiation can be understood purely as geometric bifurcations within a driven nonequilibrium statistical field.

## VI. CONCLUSION AND PERSPECTIVES

In this work, we have extended the framework of the Nonequilibrium Integrated Phase (NEIP) from a scalar field to a complex vector field, formulating the extended Integrated Field Equation (Vector-IFE) to capture macroscopic coherence and active adaptation in complex driven systems. By incorporating inertial dynamics, a small-world topological kernel, and a variational free energy minimization term, we successfully integrated statistical field theory with predictive error minimization dynamics.

Furthermore, we demonstrated that the hierarchical information flow in this system can be rigorously mapped onto a renormalization group (RG) process, where microscopic degrees of freedom are compressed into macroscopic variables. Within the replica symmetry breaking (RSB)-induced multi-valley free energy landscape, external driving forces induce the bifurcation and splitting of attractors. This structural transition provides a robust, purely physical foundation for high-order functional differentiation and global phase synchronization.

The theoretical framework established here opens several promising avenues for future research. To solidify and expand upon the current foundations, several key challenges remain as future directions:

- **Explicit Formulation of Variational Free Energy:** Deriving concrete functional forms of  $\mathcal{F}$  incorporating specific observation models, likelihoods, and priors to rigorously establish the predictive error minimization mechanism.
- **Rigorous RG Mapping and Scaling Analysis:** Formalizing the hierarchical mapping  $\mathcal{M}^{(l)}$  and momentum-space blocking transformations to explicitly extract recursion relations, fixed points, and invariant critical exponents.
- **Ablation and Causal Separation:** Systematically decoupling individual mathematical mechanisms (inertial terms, small-world topologies, and predictive drivers) to isolate their precise causal roles in phase synchronization and attractor splitting.
- **Quantitative Order Parameters and Bifurcation Protocols:** Establishing rigorous quantitative metrics for global phase coherence (such as Kuramoto-type order parameters) and systematic bifurcation analysis (Lyapunov spectra and stability criteria) to distinguish deterministic structural transitions from noise-induced switching.

Ultimately, these developments set the stage for the subsequent integration of solenoidal driving (magnetic-

like rotation terms) to explore full non-time-reversal symmetric nonequilibrium statistical fields. By bridging microscopic physical interactions with macroscopic informational order, this ongoing series of works paves the way toward a unified geometric understanding of complex, driven systems.

## DATA AVAILABILITY

The numerical code and dataset generated during this study are available from the corresponding author upon reasonable request.

## ACKNOWLEDGMENTS

The author thanks the anonymous reviewers for their constructive comments and suggestions, which helped to significantly improve this manuscript.

- 
- [1] S. Takemoto, [Systematization of the nonequilibrium integrated phase \(neip\)](#) (2026).
  - [2] T. Speck, *Physical Review E* **105**, 064601 (2022).
  - [3] K. Friston, *The free-energy principle: a unified brain theory?*, Vol. 11 (Nature Publishing Group, 2010) pp. 127–138.
  - [4] R. P. N. Rao and D. H. Ballard, *Nature Neuroscience* **2**, 79 (1999).
  - [5] J. Zinn-Justin, *Quantum Field Theory and Critical Phenomena* (Oxford University Press, 2002).
  - [6] G. Parisi, [Physical Review Letters](#) **43**, 1754 (1979).